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Responsible Digital Technology Adoption in Nigerian Secondary STEM Education: A Rights, Religious-Pluralism and Child-Safeguarding Framework

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Abstract

Digital platforms, learning analytics and generative artificial intelligence may widen access to STEM resources, support teachers and enable new forms of inquiry. They may also collect children's data, mediate speech and belief, amplify bias, expose learners to harm or displace accountable human judgment. This normative policy review examines responsible adoption in Nigerian secondary STEM education through constitutional rights, child-rights standards, data-protection law, education policy and international ethical guidance. It proposes the RIGHTS framework: Rights and best interests; Inclusion and equitable access; Governance of data and vendors; Human agency and pedagogical purpose; Trust, transparency and contestability; and Safeguarding across the technology lifecycle. Religious pluralism is treated as a rights and professional-practice issue: public education should neither impose nor disparage belief, and digital systems should support age-appropriate inquiry without profiling, stereotyping or coercion. The framework specifies a pre-procurement assessment, proportionate data controls, accessibility and offline alternatives, human review of consequential outputs, reporting and remedy pathways, and periodic exit decisions. It is a legal-ethical and policy synthesis, not legal advice, an empirical survey or a theological adjudication. Nigerian schools and authorities should obtain competent legal and safeguarding review for particular deployments. Responsible innovation is defined not by tool novelty but by demonstrated educational value, lawful and minimal data use, child participation, pluralism, safety and accountable human control.

Keywords: children's rights; digital education; religious pluralism; safeguarding; data protection; Nigeria; artificial intelligence

1. Introduction

Digital technology in education is not a neutral delivery channel. A platform decides what can be entered, what is made visible, which behaviour is measured and which responses are prioritised. When learners are children, these design choices engage duties of care and rights to education, privacy, expression, information, non-discrimination, participation and freedom of thought, conscience and religion. STEM education adds further concerns because automated tutors, simulations, assessment systems and generative AI can present uncertain outputs with scientific authority.

Nigeria's National Digital Learning Policy articulates inclusive access and a national digital-learning ecosystem, while the Nigeria Data Protection Act 2023 regulates personal-data processing. The Child Rights Act and international child-rights guidance require the best interests and evolving capacities of the child to be considered. Constitutional protection of privacy and freedom of thought, conscience and religion forms part of the broader legal setting. These sources do not reduce governance to a single consent form; they require purpose, proportionality, protection and remedy.

This review asks what an accountable adoption process should require before, during and after a digital tool is used in Nigerian secondary STEM education. It gives particular attention to religious pluralism because digital content and recommender systems may enter domains of worldview, identity and moral judgment even when the product is marketed as technical.

2. Method, Scope and Limits

The paper is a normative synthesis of Nigerian constitutional and statutory materials, the Federal Ministry of Education's digital-learning policy, the United Nations Committee on the Rights of the Child's General Comment No. 25, UNESCO guidance on AI ethics and generative AI in education, and UNICEF online-safety guidance. It translates general principles into school-level decision controls. It does not provide a judicial interpretation, replace a data-protection impact assessment or offer legal advice.

No primary empirical research was undertaken. The paper reports no prevalence of online harm, teacher attitude, student belief or technology effectiveness. Religious traditions are not compared for truth or educational value. Religious pluralism here means that learners and staff with different religious or non-religious convictions should be treated with equal dignity and should not be coerced, profiled or stereotyped through technology.

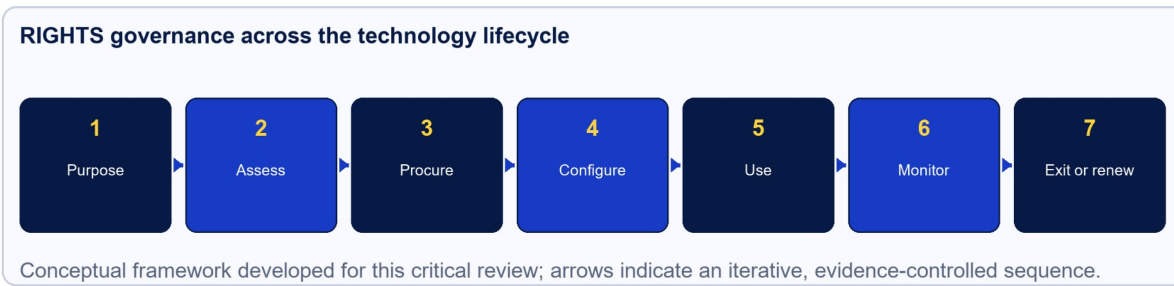

Figure 1. RIGHTS governance across the technology lifecycle.

Note. Original conceptual diagram developed for this paper; the sequence is iterative and has not been validated as a measurement scale.

3. The RIGHTS Framework

Table 1. RIGHTS principles and operational questions

Principle	Operational question	Required control
Rights and best interests	Does the use advance education while respecting the whole bundle of children's rights?	Written best-interests and proportionality assessment
Inclusion and equitable access	Who is excluded, burdened or misrepresented?	Accessibility, offline alternatives and equity testing
Governance of data and vendors	What data and third parties are necessary for the defined purpose?	Data map, lawful basis, minimisation, contract and deletion schedule
Human agency and purpose	Which educational problem requires the tool, and where must people decide?	Pedagogical use case, staff competence and human review
Trust, transparency and contestability	Can a learner understand and challenge consequential use?	Age-appropriate notice, traceability and appeal
Safeguarding	How are contact, content, conduct and commercial risks prevented and addressed?	Risk register, reporting route, response protocol and monitoring

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.

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3.1 Rights and the best interests of the child

A best-interests analysis should identify educational benefits, foreseeable harms, alternatives and the views of affected children. No right should be treated as an automatic trump card. For example, safety monitoring may support protection but become disproportionate surveillance if it collects continuous behavioural or biometric data without necessity. The Committee on the Rights of the Child (2021) emphasises that children's rights apply in the evolving digital environment and that children should be heard in matters affecting them.

Schools should distinguish mandatory institutional processing from optional learner choices. Consent is not freely given merely because a pupil clicks an acceptance box in a compulsory class. Legal basis, parental or guardian involvement, the child's evolving capacity and the authority of the education provider need case-specific assessment under applicable law and policy.

3.2 Inclusion and equitable access

Digital adoption can widen existing inequalities through device cost, data cost, electricity, language, disability access, gendered control of devices or rural connectivity. An inclusion assessment should test real conditions: shared phones, intermittent access, screen-reader operation, captioning, keyboard navigation, low-bandwidth performance and printable or offline alternatives. A learner should not receive a lower-quality curriculum solely because an inaccessible platform was selected.

Algorithmic systems require subgroup testing where lawful and statistically meaningful. Absence of a sensitive attribute in the interface does not prove absence of discrimination because location, school, language or device can operate as proxies. Small samples and intersectional groups require cautious interpretation and privacy protection.

3.3 Governance of data and vendors

The Nigeria Data Protection Act 2023 requires accountable processing of personal data. Before procurement, the institution should map data fields, purposes, actors, locations, retention, disclosures and automated decisions. Only data necessary for the educational purpose should be collected. Contracts should allocate controller and processor responsibilities, security, incident notification, subcontractors, cross-border transfers, assistance with data-subject rights, deletion or return, audit evidence and termination.

Children's data should not be repurposed for targeted advertising, unrelated product development or model training by default. High-risk processing may require a formal data-protection impact assessment. Accounts should use role-based access, strong authentication and least privilege. Logs should support security without becoming indefinite behavioural dossiers. Deletion must be verified, including backups and derived profiles where applicable.

3.4 Human agency and pedagogical purpose

A tool should enter a classroom because it supports a stated learning objective more effectively, safely or accessibly than feasible alternatives. Generative AI may help produce examples, explanations or code, but its outputs require verification against reliable sources and curricular intent. UNESCO (2023b) recommends a human-centred, age-appropriate approach, including privacy protection and ethical and pedagogical validation.

Consequential decisions—grading, discipline, admission, welfare referral or special-needs placement—should not rest on an opaque automated output. Qualified staff must understand the limitations, inspect relevant evidence, record reasons and be able to depart from the system. Learners should practise checking sources, uncertainty, assumptions and bias rather than treating fluent output as authority.

3.5 Trust, transparency and contestability

Transparency must be usable. An age-appropriate notice should explain what the system does, which data it uses, whether output is generated or recommended, who can see records, how long records remain and how to ask a question or object. Teachers require deeper documentation of model limits, configuration and escalation. Procurement teams require evidence about security, performance, accessibility and subcontractors.

A learner should be able to correct inaccurate data and challenge a consequential outcome without retaliation. Appeals need a named human, a response time and a record. Explanations should address the reasons relevant to the decision, not provide a generic claim that an algorithm is complex.

3.6 Safeguarding across the lifecycle

UNICEF groups many online concerns around harmful contact, content and conduct; commercial and data risks add another layer. School controls should address cyberbullying, grooming, self-harm content, sexual exploitation, misinformation, impersonation, coercive design and unauthorised purchases. Filters can assist but do not replace supervision, digital literacy, trusted reporting and trained response.

Staff should know what constitutes an immediate safety risk, how to preserve necessary evidence without excessive circulation, whom to notify and how to support the affected child. Learners and caregivers need clear reporting channels. A vendor's reporting button is not a substitute for the school's safeguarding responsibility.

4. Religious Pluralism in Digital STEM Learning

STEM topics can intersect with origins, reproduction, climate, health, evolution, artificial intelligence and environmental stewardship. Professional teaching should present curriculum-grounded disciplinary knowledge and the standards by which scientific explanations are evaluated. At the same time, educators should not ridicule, compel or infer a learner's belief. A distinction between explaining a scientific model and requiring personal assent should be maintained where sensitive worldview questions arise.

Digital systems can threaten pluralism through stereotyped examples, automatic religious classification, imbalanced content retrieval or prompts that invite personal disclosure. Schools should prohibit unnecessary collection of religious-belief data and review generated content for derogation, proselytisation, suppression or false equivalence. Where discussion is educationally relevant, ground rules should support evidence, respect, voluntary personal disclosure and the right not to speak for a group.

Table 2. Pluralism risks and proportionate controls

Risk	Illustrative failure	Control
Profiling	The platform infers belief from name, language, location or browsing	Prohibit unless strictly necessary and lawfully justified; test proxies
Stereotyping	Generated examples portray a faith community as anti-science	Curated prompts, source review, incident reporting and correction
Coercion	A compulsory activity solicits personal profession or rejection of belief	Assess disciplinary necessity; allow non-disclosure and neutral alternatives
Suppression	Filters block legitimate curriculum material because of religious keywords	Test over-blocking; provide teacher override and appeal

Risk	Illustrative failure	Control
Harassment	Peer collaboration exposes learners to belief-based abuse	Moderation, conduct rules, reporting and safeguarding response
False authority	AI presents a contested moral or theological claim as consensus	Label generated output; require source checking and qualified facilitation

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.

5. Lifecycle Governance

Table 3. Decision gates for responsible adoption

Gate	Evidence required	Stop or redesign trigger
Purpose	Defined learning need, alternatives and success criteria	No material educational advantage
Rights assessment	Best interests, participation, equality and privacy analysis	Disproportionate or unmitigable rights risk
Procurement	Accessibility, security, data and subcontractor evidence	Vendor refuses necessary contract or audit terms
Pilot	Bounded users, training, baseline and incident readiness	Serious harm, exclusion or unreliable performance
Deployment	Approved configuration, notices, support and human oversight	Use expands beyond assessed purpose
Monitoring	Learning value, equity, incidents, complaints and drift	Benefit not demonstrated or risk exceeds threshold
Exit or renewal	Export, deletion, continuity and fresh assessment	Lock-in, unverified deletion or obsolete safeguards

Note. Authors' synthesis unless otherwise stated; application requires context-specific evidence and professional judgment.

Governance should assign an accountable owner and include educational, safeguarding, data-protection, information-security, accessibility and learner perspectives. A central register should record tools, versions, purposes, populations, data categories, vendors, risk level and review dates. Material product changes trigger reassessment. Shadow use by staff or learners should be addressed through workable policy and approved alternatives, not denial.

Incident metrics need context: low reporting may indicate safety or fear and inaccessibility. Monitor reporting awareness, resolution time, recurring failure modes, accessibility defects, appeal outcomes, learning value and unequal effects. Public communication should describe limitations and corrective action without exposing affected children.

6. Limitations and Research Agenda

Legal duties depend on the institution, data flow, technology and current law; this paper cannot resolve every case. Policy documents express principles but do not prove implementation. Vendor evidence may be incomplete, and rapidly changing models create version drift. The framework therefore requires competent local review and documented reassessment.

Priority research should include child-participatory studies of notices and appeals, accessibility and language audits, independent evaluation of learning benefits, studies of religious and non-religious learners' experiences, and incident taxonomies that protect confidentiality. Research involving children or sensitive beliefs requires ethics approval, assent and consent arrangements, minimised data and safeguarding protocols.

7. Conclusion

RIGHTS makes responsible digital adoption a continuous governance practice. Educational value must be demonstrated alongside lawful, minimal and secure data use; equitable access; human judgment; intelligible explanation; pluralism; and safeguarding. The framework supports innovation but rejects the idea that adoption is progress by itself. Nigerian education authorities and schools should pilot deliberately, listen to children, retain accountable human control and stop deployments whose benefits do not justify their risks.

Declarations

Ethics and participant involvement

Not applicable. This review/framework used publicly available documentary and scholarly sources. It involved no human participants, animals, patient material, identifiable records or intervention.

Data availability

No primary dataset was generated or analysed. The documentary and scholarly sources used are identified in the reference list; access conditions remain those of the source publishers.

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Figure and table provenance

The conceptual figure and synthesis tables were created specifically for this review. They contain no patient, participant, survey or third-party image data.

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